

Bike Sharing Demand Analysis Using Excel

NextHikes IT Solutions | Aditi Kumar | October 2026

1. Executive Summary

I used Microsoft Excel to analyze 1,000 bike-sharing records from 1 January to 14 February 2011. I cleaned the data, analyzed demand patterns, rider behavior, calendar and weather conditions, made predictions, detected anomalies and made a dashboard.

Metric	Value
Total rentals	58,304
Average rentals per hour	58.30
Registered / Casual	91.6% / 8.4%
Peak hour — 17:00	140.20
Working / Non-working day	64.14 / 47.65
Clear / Light rain	61.67 / 33.20
First 7-day forecast	1,500.49/day

Key Takeaways

- The peak demand for rental is around **08:00** and **17:00**. The pattern of registered riders follows this peak hour.
- **Registered riders account for 91.6%** of total rentals.
- Working-day demand is higher than non-working-day demand.
- Light-rain demand is approximately **46% lower** than clear-weather demand.
- Temperature has a stronger association with **casual** than registered rentals.

Business implication: Schedule bike availability for peak hours. Weather and short-term forecasts can be useful for planning.

2. Objective, Dataset & Methodology

Objective

My objective is to understand how rental demand varies with **rider type, working days, and weather conditions**. I then utilize the results to make recommendations for how to limit bike availability.

Dataset Summary

Item	Detail
Records / period	1,000 hourly records, 1 Jan – 14 Feb 2011 (45 days)
Rentals per hour	Average 58.30; minimum 1; maximum 249; standard deviation 50.99
Registered / casual	53,383 / 4,921
Season	Winter only

Data note:

Hours with no rentals are not represented as rows. The final date, **14 February 2011**, contains only 8 hourly observations and ends at 07:00.

Methodology

Source Data → Power Query Cleaning → Merge & Append → Adding Lookup Data → PivotTables & Charts → Correlation Analysis → Forecasting → Anomaly Detection → Dashboard

Data preparation involved checking data types, validating records, checking for duplicates, adding information via lookup tables, and replacing 11 missing values with the mean.

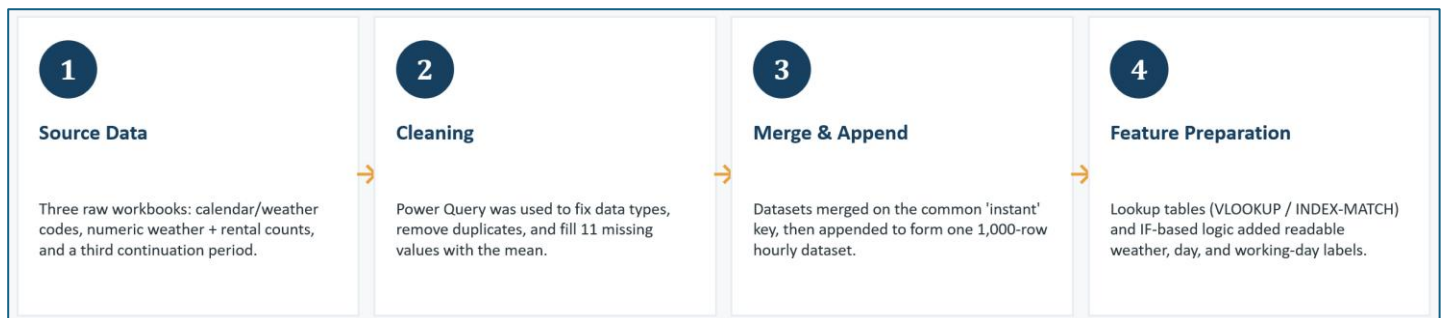


Figure 1. Data preparation workflow

3. Demand & Rider Analysis

Demand is highest during the **morning and evening peak periods**, but in the morning the demand is much lower.

Hour	Average rentals
17:00	140.20
08:00	128.29
04:00	2.16

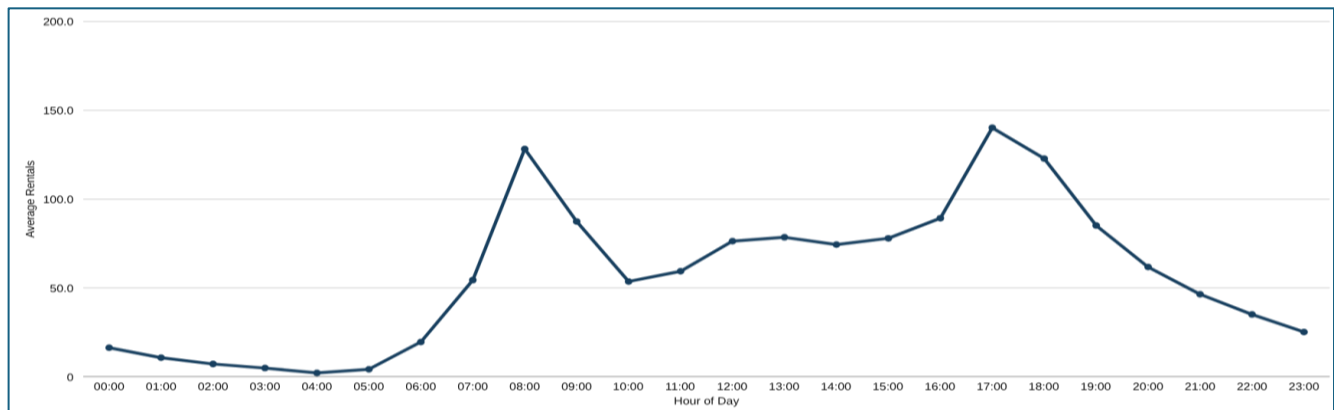


Figure 2. Average rentals by hour

Rider Type Pattern

Registered riders account for 53,383 rentals (91.6 %) and casual riders account for **4,921 rentals (8.4 %)**.

Registered riders show stronger morning and evening peaks, while casual demand is more distributed across daytime hours. Registered user demand peaks at **17:00 with 131.23 rentals**, and casual user demand peaks at **15:00 with 12.47 rentals**.

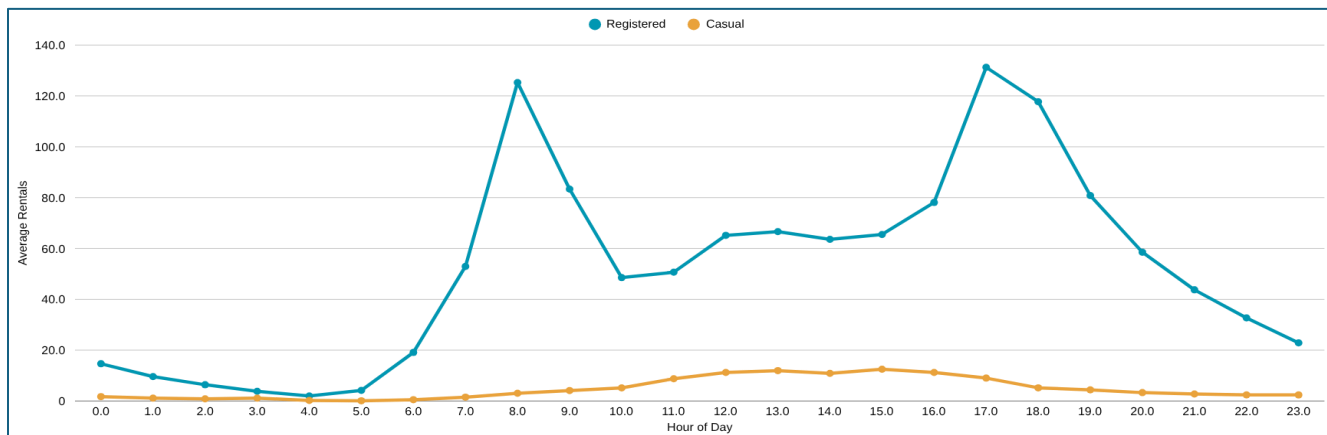


Figure 3. Registered vs casual by hour

4. Calendar Analysis

Demand is higher on **weekdays and working days** than on **weekends and non-working days**.

Comparison	Higher Average	Lower Average
Weekday vs weekend	63.34	48.09
Working vs non-working day	64.14	47.65

Thursday is the highest day of the week (**67.66**) and Saturday the lowest (**46.67**).

Holiday records average **41.67 rentals per hour** compared with 58.71 on non-holidays. However, all **24 holiday records** come from **17 January 2011.**, so this should not be interpreted as a general holiday effect.

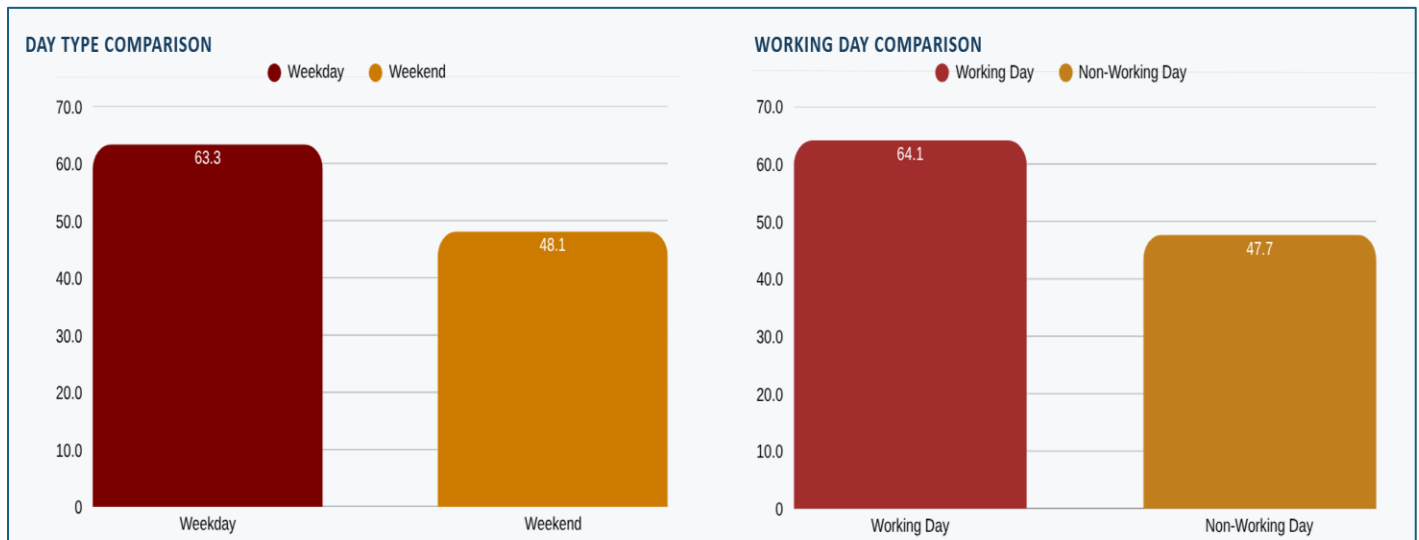


Figure 4. Weekday/weekend and working-day comparison.

5. Weather & Environmental Analysis

Light rain reduces average demand to about half of clear-weather levels (33.20 vs 61.67).

Weather	Records	Average rentals per hour
Clear	606	61.67
Cloudy	309	58.60
Light rain	84	33.20
Heavy rain	1	36.00 (not reliable)

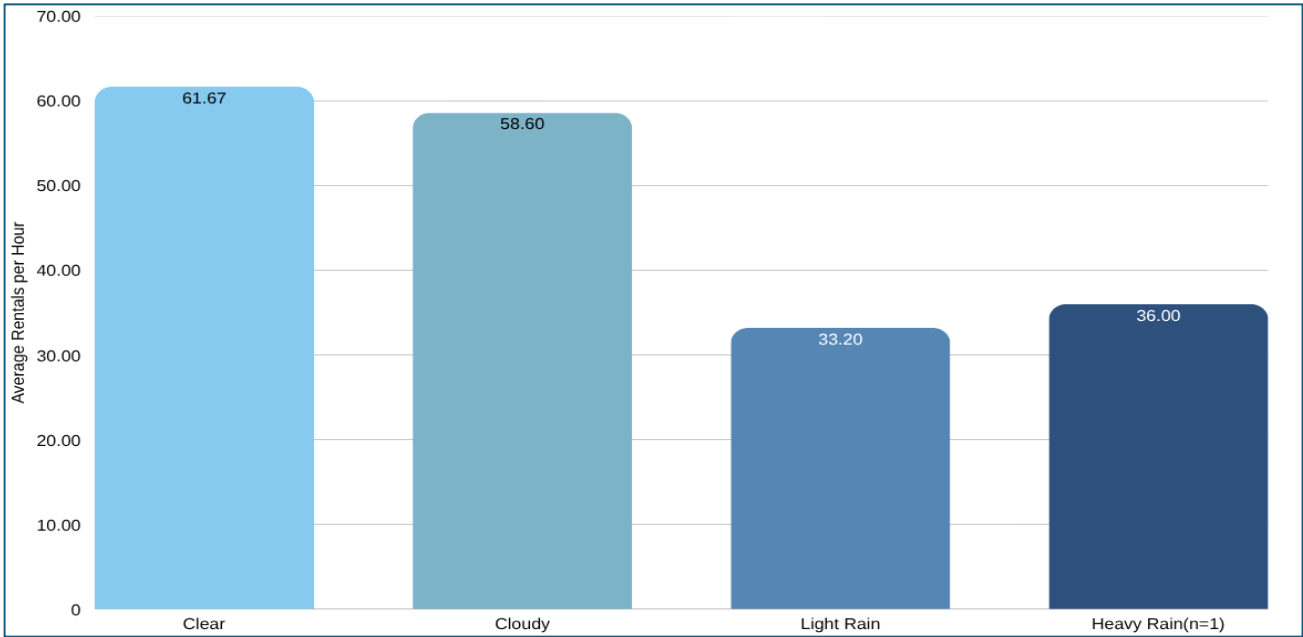


Figure 5. Average rentals by weather

Pearson correlation with rentals

Variable	Total	Casual	Registered
Temperature	+0.229	+0.470	+0.169
Apparent temperature	+0.193	+0.424	+0.138
Humidity	-0.254	-0.259	-0.229
Windspeed	+0.104	+0.129	+0.090

- Temperature has the strongest association with **casual** rentals.
- Humidity has a small negative effect on demand, while windspeed has very little effect.
- These results show associations between weather and rentals. They do not prove that weather alone causes demand to change.

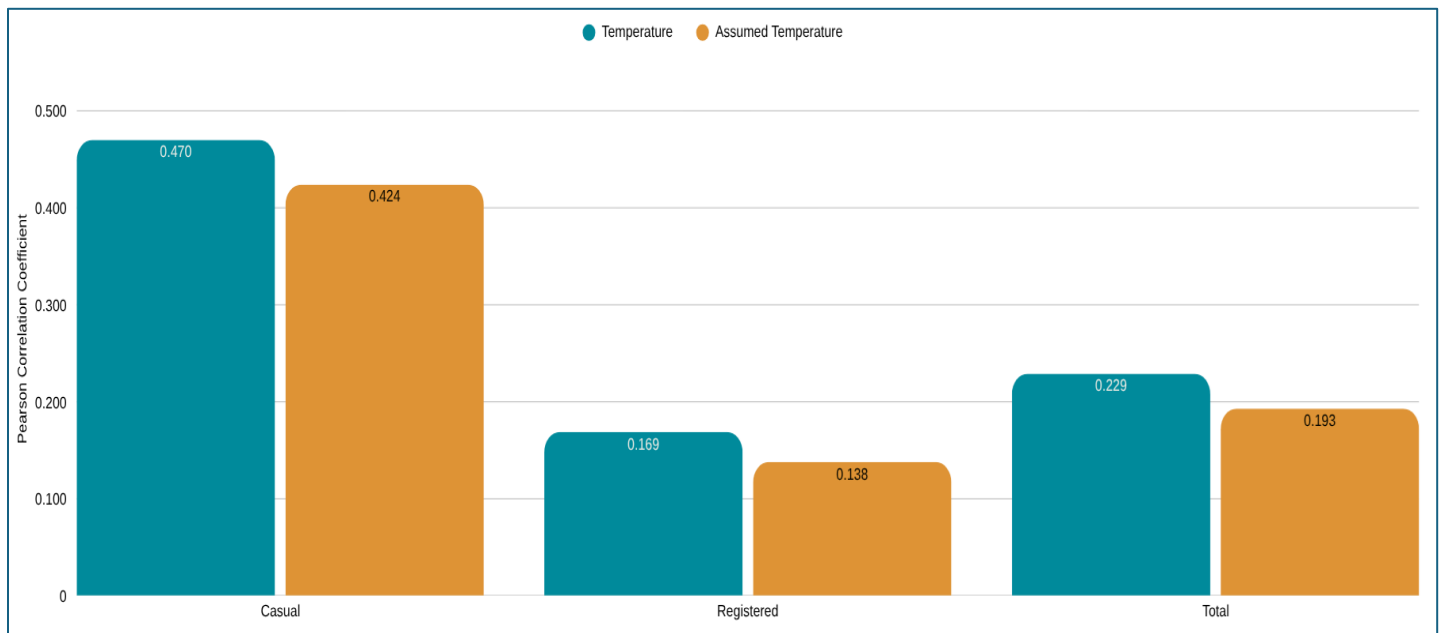


Figure 6. Correlation chart

6. Temperature & Forecast

Average rentals increase across the temperature bands and reach their highest level in the **0.3–0.4 band at 93.88 rentals per hour**. Demand then falls to 59.48 rentals per hour in the 0.4–0.5 band.

Temperature values are normalized from **0 to 1**, so these bands represent normalized temperature and not direct temperature units.

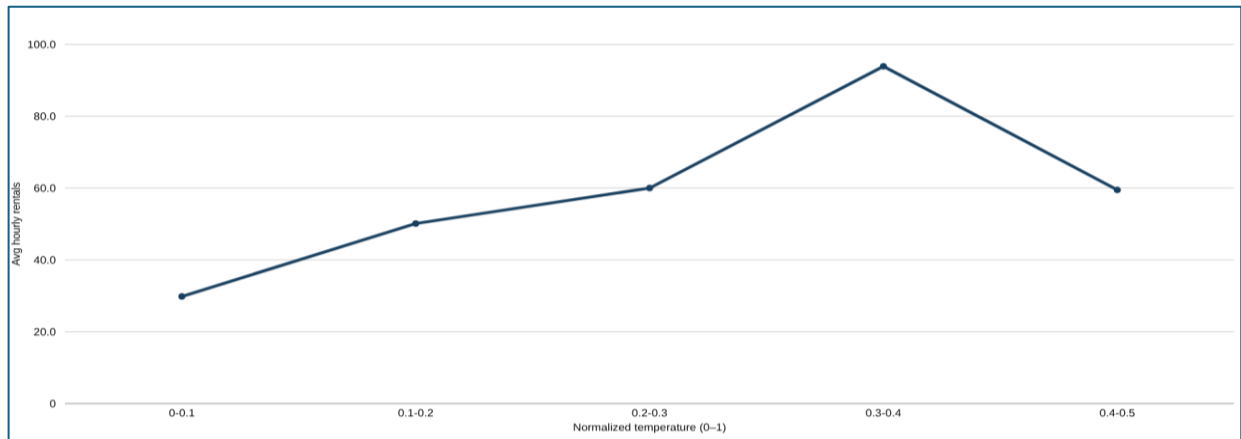


Figure 7. Demand by temperature band

Excel's FORECAST.ETS projected daily demand for 15 Feb – 7 Mar 2011 (21 days).

Forecast measure	Rentals per day
Historical average (45 days)	1,295.64
First 7 forecast days	1,500.49
Full 21 forecast days	1,543.28

The forecast suggests an average of about 1,500 rentals each day over the seven days. This is an estimate, not a guaranty and actual demand may vary.

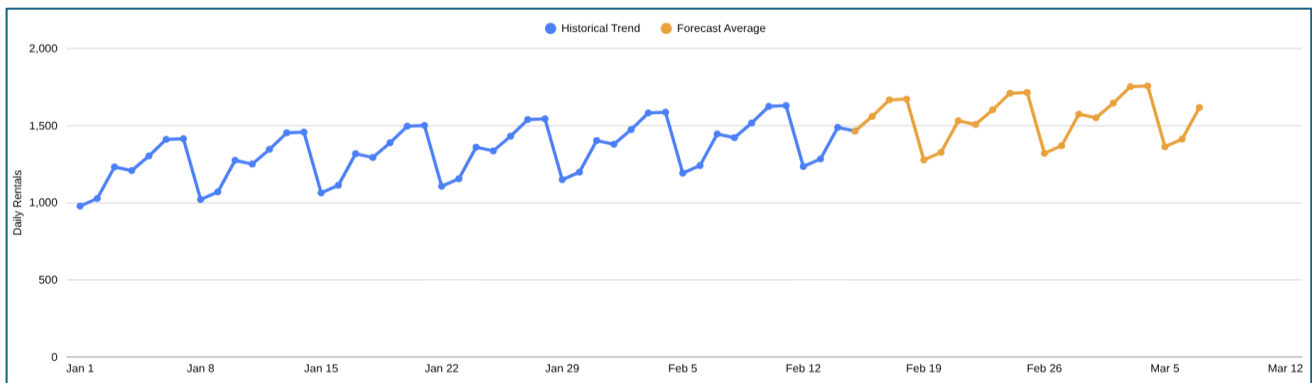


Figure 8. Historical and forecast demand

7. Anomaly Detection & Dashboard

The **IQR method** was used to identify unusually high or low daily rental totals.

Metric	Value
Q1	1,005
Q3	1,562
IQR	557
Lower Threshold	169.5
Upper Threshold	2,397.5

One of the 45 days was flagged: **14 February 2011**(151 rentals).

However, only **8 hourly observations** are available for that date, ending at 07:00. Therefore, the result is best treated as a **partial-day/data-availability** and should not be interpreted as confirmed drop in demand.

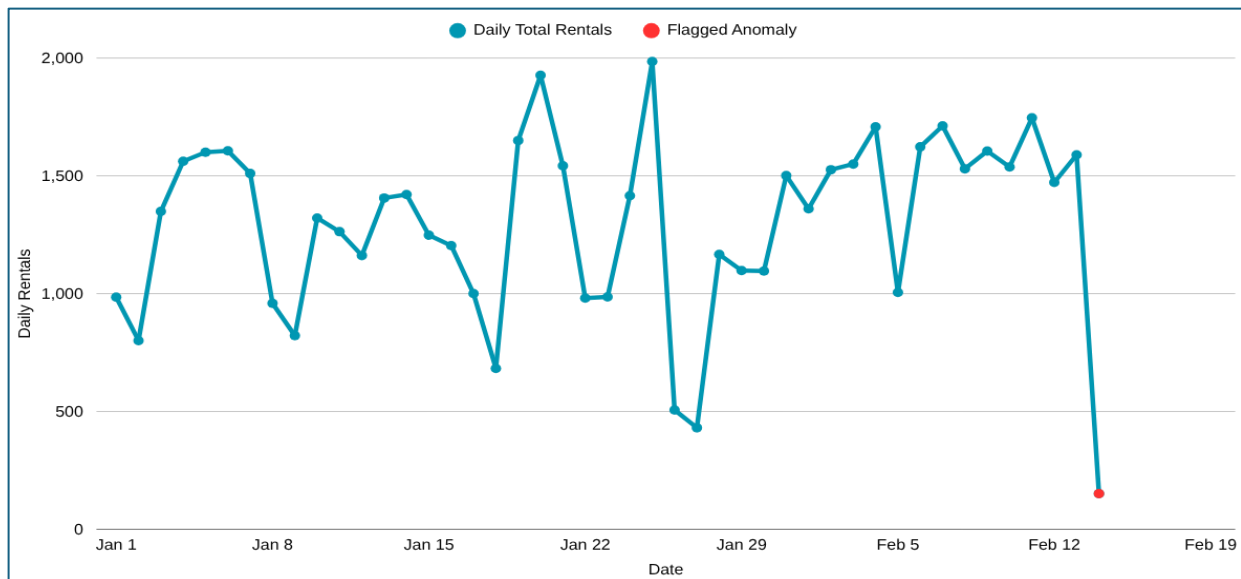


Figure 9. Daily rentals with flagged anomaly

8. Excel Dashboard

The dashboard combines KPIs, hourly, rider, weather and forecast charts, with slicers and a timeline for filtering. Power Query and PivotTables make updating reports with new data easier.



Figure 10. Excel Dashboard

9. Findings, Recommendations & Conclusion

Key Finding	Recommendation
91.6% of rentals are by registered riders. Most active in spikes at 08:00 and 17:00.	Always check on bike availability and make adjustments before peak times when Registered riders are particularly active.
Weekend demand is about 24% lower than weekday demand.	Adjust availability and process planning based on whether it is a weekend or a weekday.
Demand in light rain is 46% below demand on clear weather days.	Weather forecasts can be a supporting information for the planning process.
Temperature is linked with the casual riders.	Design capacity for riders around weather conditions.
The rentals are averaging about 1,500 per day, over a 7-day average.	Use the forecast for planning in the short range. Keep it updated with fresh data regularly.
Only 45 winter days are available.	Request longer, seasonal data in order to confirm patterns.

Limitations

- Winter-only data covering 45 days; 14 Feb is a partial day.
- Heavy rain has one observation; the holiday result comes from one holiday.
- Correlation does not establish causation.
- The forecast is based on historical data and so the predictions may not be in line with actual demand.

Conclusion

This project shows that bike rental demand varies with the time-of-day rider type, working-day status and weather conditions. Registered riders make up rentals and show clear morning and evening peaks while casual rentals have a stronger relationship with temperature.

Excel was used throughout the process from data cleaning and analysis to forecasting, anomaly detection and dashboard creation. The findings can help with bike availability and short-term planning. More data, from different seasons is needed to confirm the patterns and improve future predictions.